

D7054E Lab 1

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Abstract

A **Dataset** class was created with all the necessary methods, including data cleaning and statistical calculations. Two subclasses, **Happiness_Dataset** and **Covid_19_Dataset**, extend the **Dataset** class and inherit its methods. The program identifies numeric columns and cleans them. After that, it computes statistical measures and searches for a specific country or region. A UML class diagram is included to illustrate the system design.

Introduction

Python was used in this lab to create a dataset processing system. The objective was to design a base class capable of loading and processing CSV files, and then use inheritance to support specific datasets.

To handle data loading, row cleaning, numeric column selection, and statistical calculations such as mean, variance, standard deviation, minimum, and maximum, a parent class named **Dataset** was implemented. Two subclasses, **Happiness_Dataset** and **Covid_19_Dataset**, were created to represent specific datasets while inheriting the required functionality from the **Dataset** class.

The program processes both datasets, displays their structure, computes statistics for numeric columns, and performs a simple search operation. A UML class diagram is included to illustrate the system design.

1 Methodology

In this lab, Python was used to build a dataset processing system using an object-oriented approach. Two CSV datasets were used: the WHO COVID-19 dataset and the World Happiness dataset. The system was designed using inheritance, where a base class handles common functionality and subclasses represent specific datasets.

Class Design

To reuse common functionality such as loading CSV files, cleaning rows, selecting numeric columns, and computing statistical measures, a parent class named **Dataset** was implemented. Two subclasses, **HappinessDataset** and **Covid_19_Dataset**, inherit from the base class and only override the dataset name. This design enables code reuse and demonstrates the use of inheritance and basic polymorphism.

Data Structures

The program uses the following Python data structures:

- A list of dictionaries to store dataset rows.
- A list to store column names.
- A dictionary to store statistical results such as mean, variance, and standard deviation.

Data Cleaning

During the loading process, basic data cleaning was performed. Whitespace and special characters were removed from column names. Each row was processed to remove extra spaces from string values. Values were converted to numeric types using a helper function that ignored missing or invalid values.

Processing and Statistics

The application automatically identifies numeric columns. It calculates the mean, variance, standard deviation, minimum, and maximum values for each numeric column. In addition, a simple search function was implemented to locate rows matching a given country name.

2 Results

The program was executed on both the World Happiness and WHO COVID-19 datasets. The system calculated statistical measures such as mean, variance, standard deviation, minimum, and maximum values, displayed the column names, and identified the numeric columns.

The application searched for the country “Finland” in the Happiness dataset and returned the corresponding row. It also searched for “India” in the COVID-19 dataset and successfully found the matching record.

These results confirm that the application correctly loads the datasets, identifies numeric columns, computes statistical measures, and performs the required search operations. Figure 1 shows the output of the `Covid_19_Dataset`, and Figure 2 shows the output of the `Happiness_Dataset` after running the application. Figure 3 shows the UML class diagram of the implemented system.

```
[Deaths - newly reported in last 7 days]
mean      = 15153.8
variance  = 987952560.8444443
std       = 38132.251174521367
min       = 978.0
max       = 98817.0

[Deaths - newly reported in last 24 hours]
mean      = 2507.4
variance  = 25218753.60000001
std       = 5021.827715085416
min       = 79.0
max       = 16465.0

Search: Name = India
Matches found: 1
{'Name': 'India', 'WHO Region': 'South-East Asia', 'Cases - cumulative total': '10701193', 'Cases - cumulative total per 1 million population': '7754.46', 'Cases - newly reported in last 7 days': '90310', 'Cases - newly reported in last 24 hours': '11666', 'Deaths - cumulative total': '153847', 'Deaths - cumulative total per 1 million population': '111.48', 'Deaths - newly reported in last 7 days': '978', 'Deaths - newly reported in last 24 hours': '123', 'Transmission Classification': 'Clusters of cases'}
```

Figure 1: Output of the `Covid_19_Dataset`

```
[Generosity]
mean      = 0.27410000000000007
variance  = 0.0031294333333333336
std       = 0.05594133832268704
min       = 0.153
max       = 0.354

[Perceptions of corruption]
mean      = 0.31899999999999995
variance  = 0.0078962222222222225
std       = 0.08886068997156293
min       = 0.118
max       = 0.41

Search: Country or region = Finland
Matches found: 1
{'Overall rank': '1', 'Country or region': 'Finland', 'Score': '7.769', 'GDP per capita': '1.34', 'Social support': '1.587', 'Healthy life expectancy': '0.986', 'Freedom to make life choices': '0.596', 'Generosity': '0.153', 'Perceptions of corruption': '0.393'}
```

Figure 2: Output of the `Happiness_Dataset`

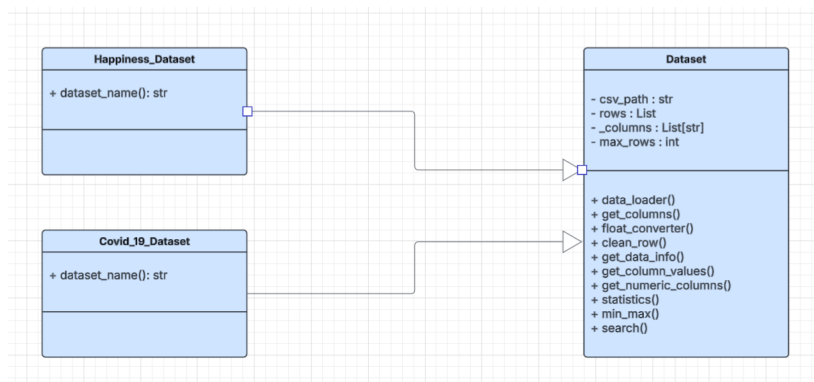


Figure 3: A UML diagram to illustrate the system design

3 Discussion

Both datasets were successfully loaded by the application, and the first 10 rows were processed as required. The statistical measures (mean, variance, standard deviation, minimum, and maximum) were computed correctly, and the numeric columns in each dataset were properly identified.

For the Happiness dataset, the calculated values were consistent with the observed data ranges, indicating that the numeric detection and statistical functions operated correctly. Similarly, for the COVID-19 dataset, the program correctly identified large numeric columns, such as cumulative cases and deaths, and computed their statistics.

The search functionality also performed as expected. When searching for specific countries (Finland in the Happiness dataset and India in the COVID-19 dataset), the correct rows were returned.

Overall, the implemented methods for data loading, cleaning, numeric column detection, statistical analysis, and searching operated correctly on both datasets.

4 Conclusion

The results showed that the implemented approach works consistently across different datasets, which demonstrates the benefit of using inheritance for reusable data processing logic. This lab helped in understanding how object-oriented design can simplify data handling and make the code more modular and reusable.